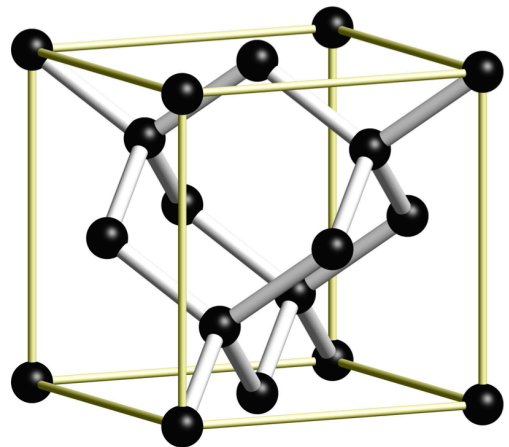


GEORGIA INSTITUTE OF TECHNOLOGY
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
ECE 3040 Final Practice Test

Contents: **15 Questions (50 points)**
3 Problems (60 points)

1. **(3 points)** For the lattice structure shown on the right

(a) (1 point) What type of lattice is this? Give an example material for this lattice structure?



(b) (1 point) What is the number of nearest neighbors for the atoms in this crystal lattice?

(c) (1 point) How many atoms are there in the unit cell?

2. **(3 points)** What is the probability of an energy state with energy $E_F + 2kT$ being occupied by an electron? What is the probability of a state with the same energy being occupied by a hole?

3. **(3 points)** Consider three intrinsic semiconductor materials with the same effective density of conduction and valence band states. The bandgaps of these semiconductors are such that $E_{G1} > E_{G2} > E_{G3}$. Please qualitatively compare the equilibrium concentration of carriers (n_i) in these semiconductors,

(a) (1 point) at room temperature.

(b) (1 point) at 0K

(c) (1 point) at 600K

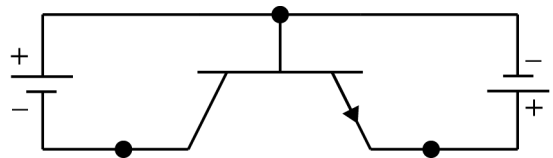
4. **(3 points)** A silicon wafer is simultaneously doped with both 10^{15} cm^{-3} boron and 10^{12} cm^{-3} phosphorus atoms.
- (a) (1.5 points) Calculate the electron and hole carrier densities at room temperature, assuming complete ionization ($n_i = 10^{10} \text{ cm}^{-3}$).
- (b) (1.5 points) Roughly sketch (no calculations necessary) the energy band diagram, properly showing the locations of the Fermi energy, intrinsic Fermi energy, conduction and valence band energy levels.
5. **(3 points)** Describe the Auger recombination process in words. Also, please draw a schematic showing the process on the energy-band diagram to support your description.

6. **(3 points)** Imagine an **n^+p junction**. Which current component (electron or hole, drift or diffusion) is dominating the current flow through the junction under following conditions? Justify your answer.

(a) (1.5 points) Under forward bias.

(b) (1.5 points) Under reverse bias.

7. **(3 points)** Identify the type of BJT and mark the emitter, base and collector terminals. Assuming $N_E \gg N_B > N_C$, sketch the energy band diagram. In your sketch clearly indicate emitter, base and collector regions and label E_C , E_V , E_i and E_F .



8. **(3 points)** For the BJT in question 7, roughly sketch the **total minority carrier** concentration in the emitter, base and collector regions. Please indicate the minority carrier type and equilibrium concentrations, mark the emitter, base and collector in your sketch, and assume $W/L_B \ll 1$.
9. **(6 points)** Sketch the **energy band diagram** and **charge diagram** of an ideal MOS (assume $\Phi_M = \Phi_S$) structure for the following conditions. In your energy band diagram, include E_F in the metal, E_C and E_V in the oxide, and E_C , E_i , E_F and E_V in the semiconductor. Also indicate the polarity of the potential difference between the metal and the semiconductor for each case (i.e., $V_G > 0$ or $V_G < 0$).
- (a) (3 points) p-type semiconductor **under inversion**.

(b) (3 points) n-type semiconductor **under accumulation**.

10. **(3 points)** Consider an MOS capacitor with a p-type Si substrate doped with 10^{17} cm^{-3} acceptor atoms at room temperature. Under delta depletion approximation, calculate the potential difference across the semiconductor

(a) (1 point) For accumulation biasing.

(b) (1 point) At the onset of inversion.

(c) (1 point) For inversion biasing.

11. **(3 points)** Sketch a cross-section of a **p-channel MOSFET** beyond the **pinch-off point (in saturation)**, properly labeling the key regions (and their doping type) and the terminals. Include the depletion region and the inversion layer in your sketch.
12. **(3 points)** How are the I_D - V_{DS} characteristics of a **short channel** and **long channel** MOSFET different? Please indicate the underlying physical reason. Which parameter of the MOSFET's small signal equivalent circuit is affected by this difference?

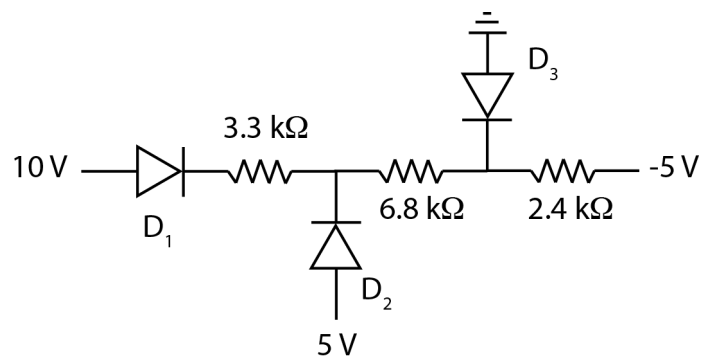
13. **(5 points)** Which of the following statements are true and which are false?

- (a) The electric field inside the gate insulator in an ideal MOSFET is constant.
- (b) There is a current flow between the gate and the source in NMOS when biased in accumulation.
- (c) A p-doped Si substrate has greater work function compared to an n-doped Si substrate.
- (d) A PMOS has n^+ -type source and drain and a p-type substrate.
- (e) In a PMOS, the drain has higher potential than the source.

14. **(3 points)** Assume you want to build a single transistor amplifier using a MOSFET. Which terminals can you use for input? Which terminals can you use for the output? Justify your answer.

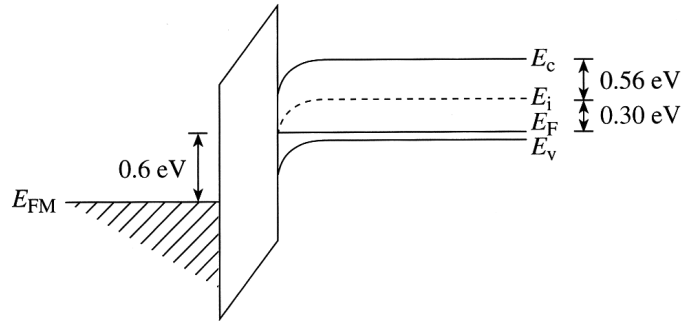
15. **(3 points)** Draw small signal models of both BJT (active-mode) and MOSFET (saturation). Include the output resistances. Clearly indicate the terminals. What is the main difference between them?

Problem 1 (10 points): Find the states (forward or reverse biased) and Q-points for the diodes in the circuit using the constant voltage drop model with $V_{on} = 0.7\text{ V}$.



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Problem 2 (25 points): Shown below is a energy band diagram for an ideal silicon MOS capacitor (i.e. the metal workfunction equals the semiconductor workfunction) operated at $T = 300$ K with $V_G \neq 0$. Note that $E_F = E_i$ at the Si-SiO₂ interface.



- (2 points) Extract the Fermi potential ϕ_F in [V] and the energy difference $E_F - E_{i,bulk}$ in [eV] far away from the Si-SiO₂ interface.
- (4 points) Calculate the carrier concentrations n and p in the quasineutral regime and determine the doping concentration of the substrate, assuming a single dopant type.
- (4 points) Calculate the resistivity of the semiconductor substrate far away from the Si-SiO₂ interface. Assume $\mu_n = 1360$ cm²/Vs and $\mu_p = 460$ cm²/Vs.
- (3 points) Extract the surface potential ϕ_s .
- (4 points) Extract the applied gate voltage V_G (including its sign)?
- (4 points) Calculate the oxide thickness x_o .
- (4 points) Calculate the threshold voltage V_T , i.e. the gate voltage required to reach the onset of inversion.

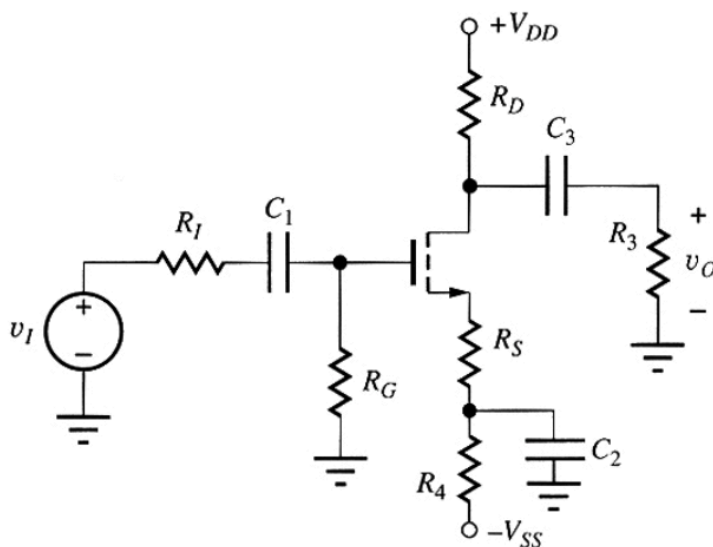
You might need: $K_O = 3.9$, $K_S = 11.7$, $\epsilon_0 = 8.85 \cdot 10^{-14}$ F/cm, $q = 1.6 \cdot 10^{-19}$ C, $kT = 0.026$ eV at room temperature, $E_g = 1.12$ eV, $n_i = 10^{10}$ cm⁻³.

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Problem 3 (25 points): You are investigating the amplifier shown below with a MOSFET transistor operated in saturation mode bias. $R_I = 2\text{ k}\Omega$, $R_G = 500\text{ k}\Omega$, $R_D = 6\text{ k}\Omega$, $R_S = 2\text{ k}\Omega$, $R_4 = 3\text{ k}\Omega$ and $R_3 = 15\text{ k}\Omega$, $V_{DD} = 5\text{ V}$, $V_{SS} = 5\text{ V}$, $V_T = 2\text{ V}$, $K_n = 600\text{ }\mu\text{A/V}^2$ and $\lambda = 0$.

- (1 point) Which single-transistor amplifier type is represented by this circuit?
- (4 points) Draw the dc equivalent circuit of the amplifier.
- (6 points) Calculate the Q-point (I_D , V_{DS}). Verify that the MOSFET is indeed saturation mode biased.
- (3 points) Calculate the transconductance g_m , the input resistance r_π and the output resistance r_o of the MOSFET.
- (5 points) Draw the ac equivalent circuit of the amplifier including the small signal equivalent circuit for the MOSFET.
- (6 points) Using the small signal equivalent circuit, extract the signal source voltage gain v_o/v_i of the amplifier.



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